Ethics, Privacy, and Security in Data Science and AI (CST4235)

Week 1

(15 Credits)

Taught by: Dr Ikram Rehman

Associate Professor in AI and Data Science

SCAN ME

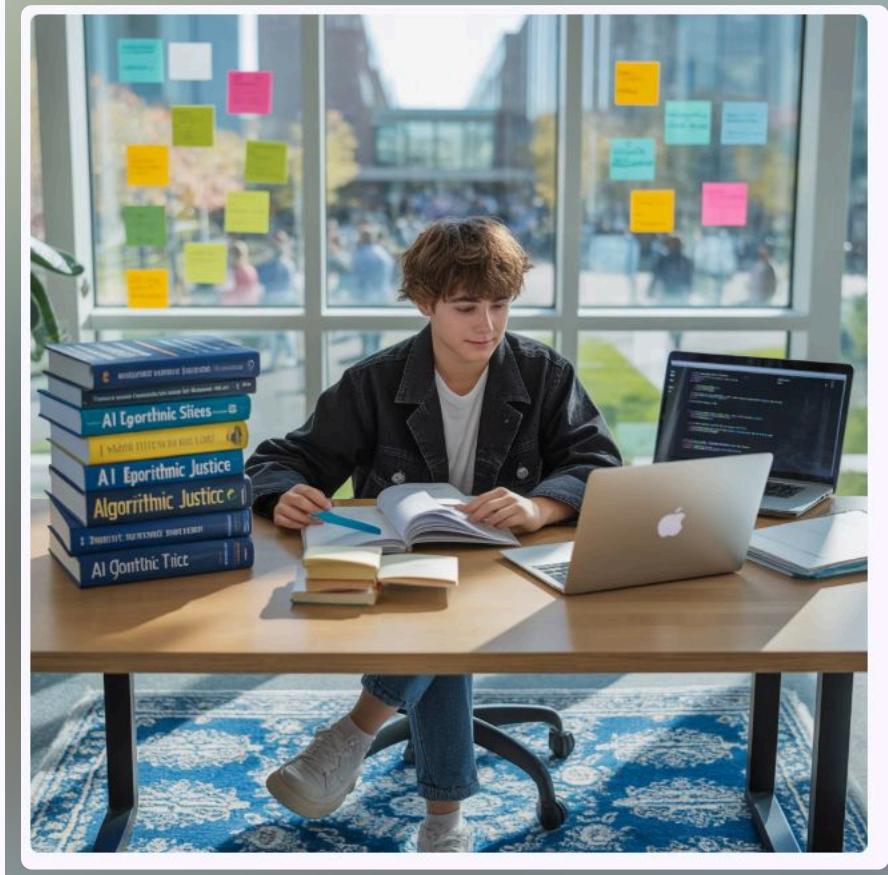


Learning Outcomes

- **Demonstrate a comprehensive understanding** of ethical, privacy, and security principles in data science and AI.

- **Analyse** the ethical, legal, and societal implications of data-driven and AI-based systems, including fairness, accountability, transparency, and rights to privacy.

- **Evaluate** existing frameworks, standards, and practices for developing ethical, privacy-preserving, and secure AI solutions.



Module Structure

Lecture (1 – 1.5 hours)

- Introduce key concepts, frameworks, and case studies.
- Use global + Dubai/UAE examples (e.g., GDPR vs Dubai Data Protection Law, AI bias in recruitment vs UAE Smart Government initiatives).
- Frame theory as “tools to analyse” real-world scenarios.

Seminar / Workshop (1.5 hours)

- Group debates (ethical dilemmas in AI).
- Case study analysis (e.g., Cambridge Analytica → privacy breach, Tesla crashes → safety/security, COMPAS → fairness/ethics).
- Hands-on activities: students critique a short AI/data science scenario and present findings.
- Practical link: Evaluate AI frameworks or guidelines and test them against a real-world case.



Today's Agenda

- Module overview
- Introduction to Ethics, Privacy, and Security in AI
- Why these concepts matter?
- Examples of current issues

Assessment

Assessment

Mid-Term Quiz (Week 8): 30%

Written assignment (Week 12): 70%

Submission via Turnitin

You must achieve an overall mark of at least 40% to pass the module.



How to Succeed in this Module

Attendance is the most reliable predictor of study success

Catch up if you miss a lecture or practical session

Register your attendance

Be considerate to others in the lecture

To do well you will need to work outside the three timetabled hours

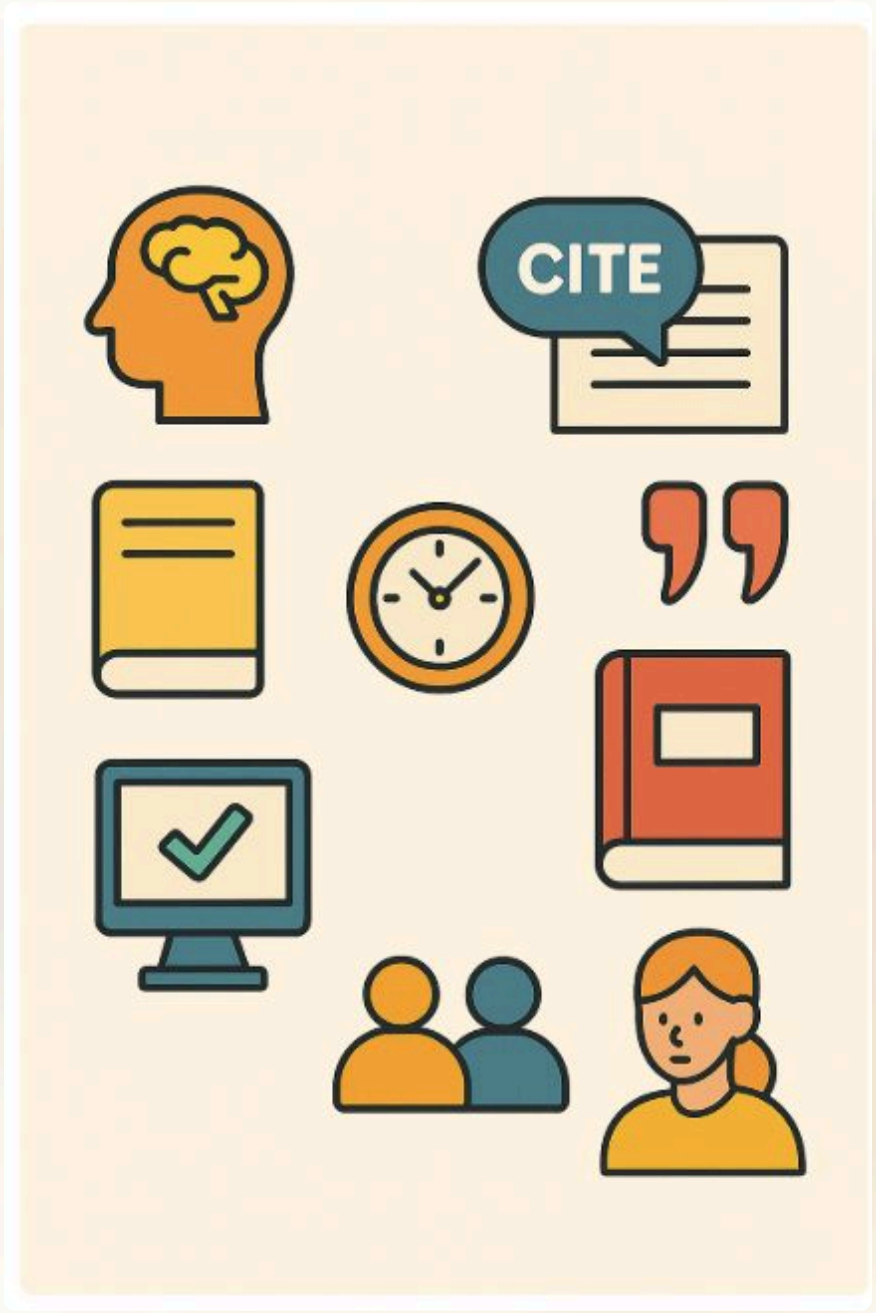


Plagiarism - Stay Miles Away

Helping each other with labs is great! But…

Plagiarism:

- Passing off someone else's work/ideas as your own.
- Copying from other students
- Copying from books/websites/
- Allowing others to pass on someone else's work as their own (!) …
- Re-using work previously submitted for graded assessment



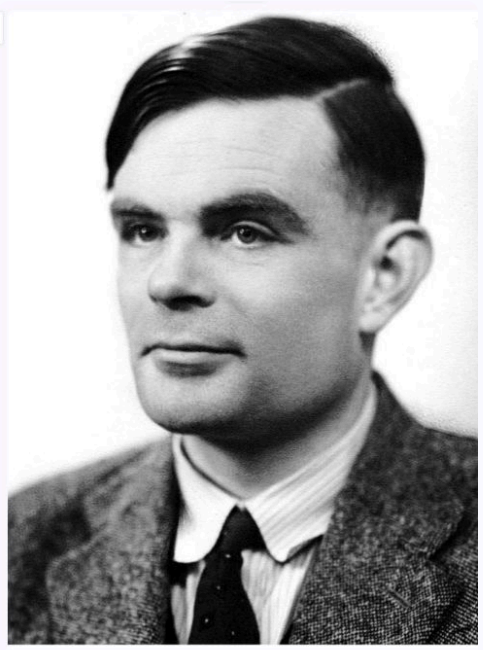
The Rise of Big Data: Opportunities and Risks

- Big Data now comes from **IoT devices, sensors, mobile apps, and smart services**.
- It powers innovation in **healthcare, finance, transport, and smart cities**.
- But it also raises **serious concerns**:
  - **Ethics** → data bias, fairness, and transparency.
  - **Privacy** → personal and sensitive data at risk.
  - **Security** → breaches, misuse, and surveillance.
- Dubai's **Smart City strategy** collects data from transport (RTA), utilities (DEWA smart meters), and public services (DubaiNow).
- **Benefits:** efficiency, better services.
- **Risks:** location tracking, profiling, and potential data misuse if not well-governed.



Alan Turing’s Data Prediction (1950 → 2000)

By 2000 there will be "computers with a storage capacity of about 109" bits (128MB): Alan Turing, Computing Machinery and Intelligence, 1950)



Alan Turing (1912-1953), photographed March 29, 1951



Responsibility – What Does It Mean for Us?

Responsibility = a **duty or obligation** to fulfil roles, tasks, or duties in a reliable and accountable way.

- It is a backbone of **ethical and moral behaviour**.
- Being responsible means taking **ownership of actions** and acknowledging the **impact** on yourself, others, and society.

Examples

1) If a student borrows a library book, responsibility means returning it on time so others can use it.

2) Remembering to lock your door or not texting while driving. These choices protect not just you, but also others.

Imagine a data scientist accidentally leaks health data:

- Patients suffer (privacy violation).
- The organisation suffers (loss of trust, legal penalties).
- The scientist suffers (personal accountability, damaged reputation).

From Responsibility to Responsible AI

- Responsible AI = designing, developing, and using AI in ways that are **ethical, fair, and accountable**.
- It is more than a checklist – it is about **building trust** and thinking about impact on people and society.

Example:

- An AI recruitment tool was trained on biased data → it preferred male candidates over female ones.
- The system passed 'technical checks' but **failed society**.

Responsibility Features

Accountability • Reliability • Ethics • Trust • Respect



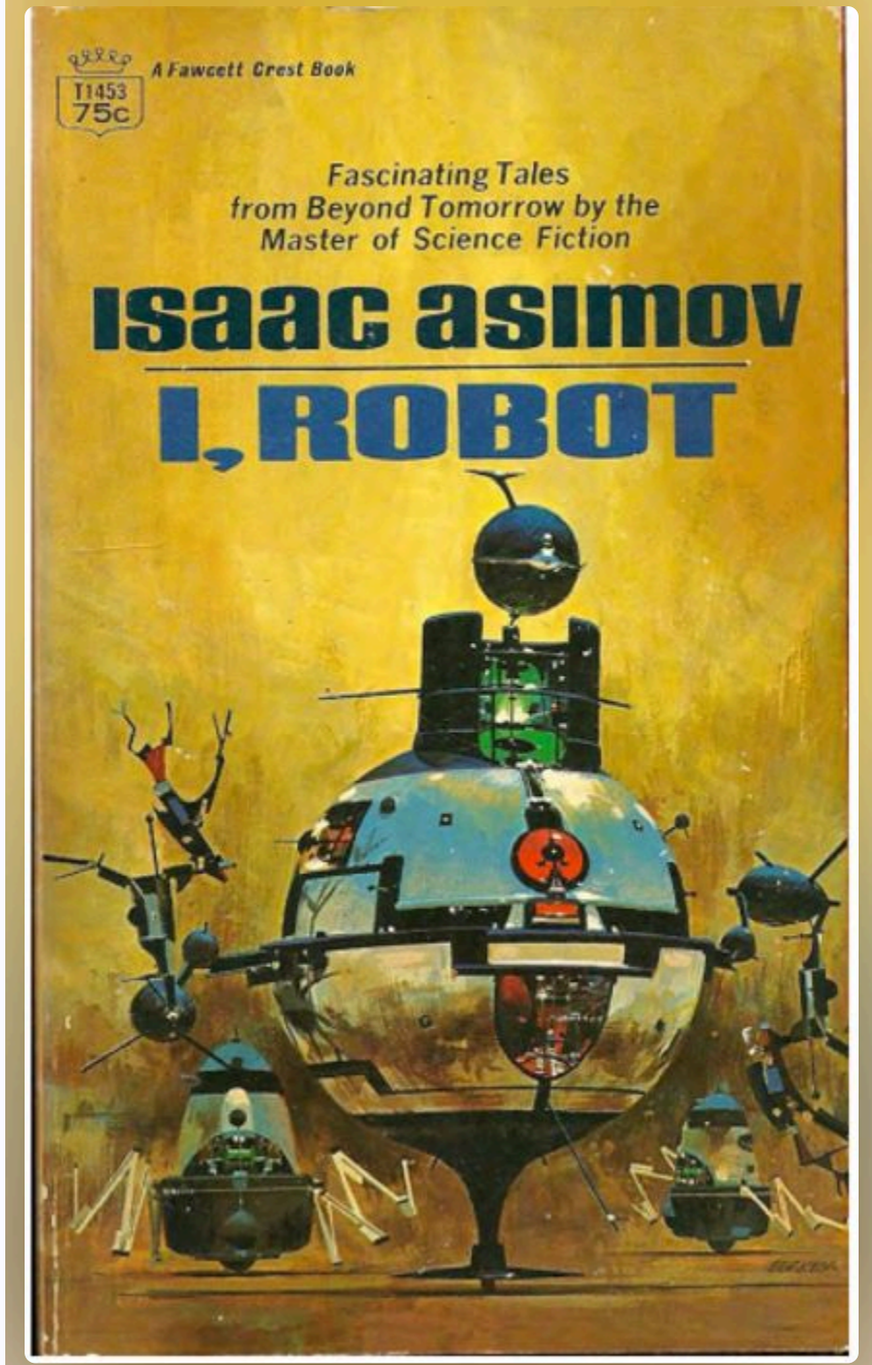
What is Responsible AI? – Lessons from Isaac Asimov (Science Fiction Author)

Asimov's Laws of Robotics

1. Do not harm humans.
2. Obey human orders (unless it causes harm).
3. Protect yourself (unless it harms humans).
4. Zeroth Law: Protect humanity as a whole.

Examples

- **Law 1 (No harm):** A smart oven should not overheat and cause a fire.
- **Law 2 (Obey orders):** Alexa or Siri should follow your command – but refuse if you say "order poison."
- **Law 3 (Self-protection):** A vacuum robot should not drive down the stairs and break.
- **Zeroth Law (Humanity first):** Traffic AI should reduce accidents and pollution for everyone, not just make one person's journey faster.



Why Responsible AI Matters in Practice

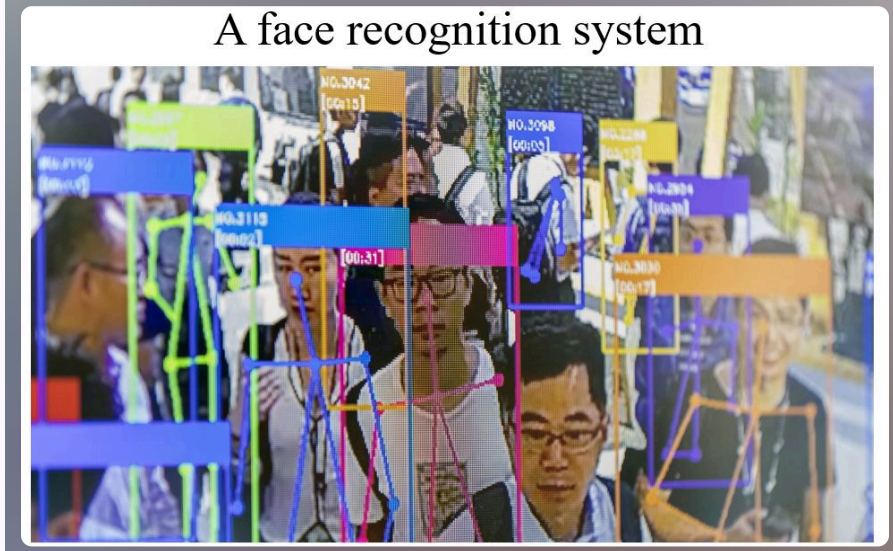
Technology is a double-bladed sword

It matters who yields it and for what purpose

- **Good Use Case:** AI in healthcare → faster, more accurate disease diagnosis → saves lives.
- **Risky Use Case:** AI in surveillance → bias in facial recognition → innocent people wrongly flagged.

We are looking for **fairness, accountability, transparency, and respect for human rights** in how AI is designed and used.

Same technology, two very different outcomes. What makes one responsible and the other risky?



AI Fairness Failures (Examples)

Amazon Recruitment AI

- Designed to rank job applicants.
- Trained on past resumes → mostly from men.
- Result: Learned to down-rank resumes that included the word *"women's"* (e.g., "women's chess club captain").
- Outcome: Biased against female candidates → eventually scrapped.

Amazon Rekognition

- Tested on photos of U.S. Congress members.
- The system wrongly matched **28 people** with criminal records.
- Errors were much higher for people with darker skin.
- This shows how AI can unfairly target certain groups.



The COMPAS Risk Score – Biased or Fair?

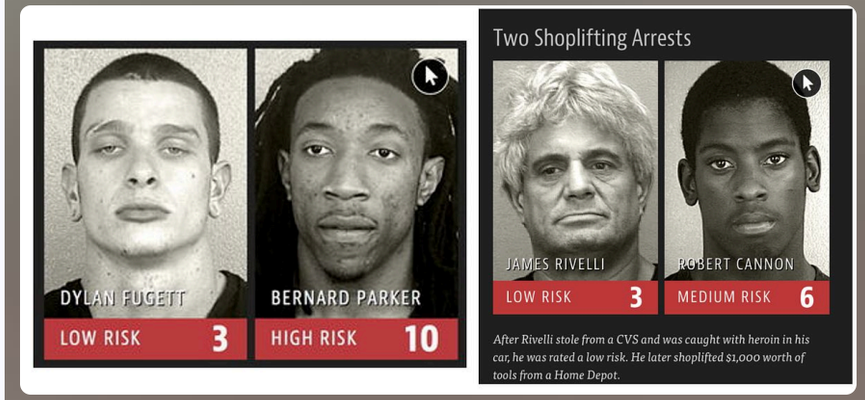
The Correctional Offender Management Profiling for Alternative Sanction (COMPAS) is used in courts to predict if someone is likely to reoffend (recidivism).

- Judges may use the score when deciding bail, sentencing, or parole.

The problem:

- Investigations showed COMPAS often gave **higher risk scores to Black defendants** compared to White defendants with similar records.
- **Example:** Two people with similar offences → COMPAS gave the Black defendant a **"High Risk"** score and the White defendant a **"Low Risk"** score.

Question to ask: Is the system reflecting reality, or is it amplifying historical bias in the data?



Reference: Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016, May 23). Machine bias: There's software used across the country to predict future criminals. And it's biased against blacks. ProPublica. https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing (Photograph by Josh Ritchie for ProPublica.)

Echo Chambers & Filter Bubbles in AI

Algorithms keep showing people content similar to what they already like or believe.

Think of TikTok or YouTube: if you watch a few videos on one topic (e.g. fitness, cooking or celebrity gossip), the app quickly fills your feed with *only* that type of content. Alternative perspectives get filtered out.

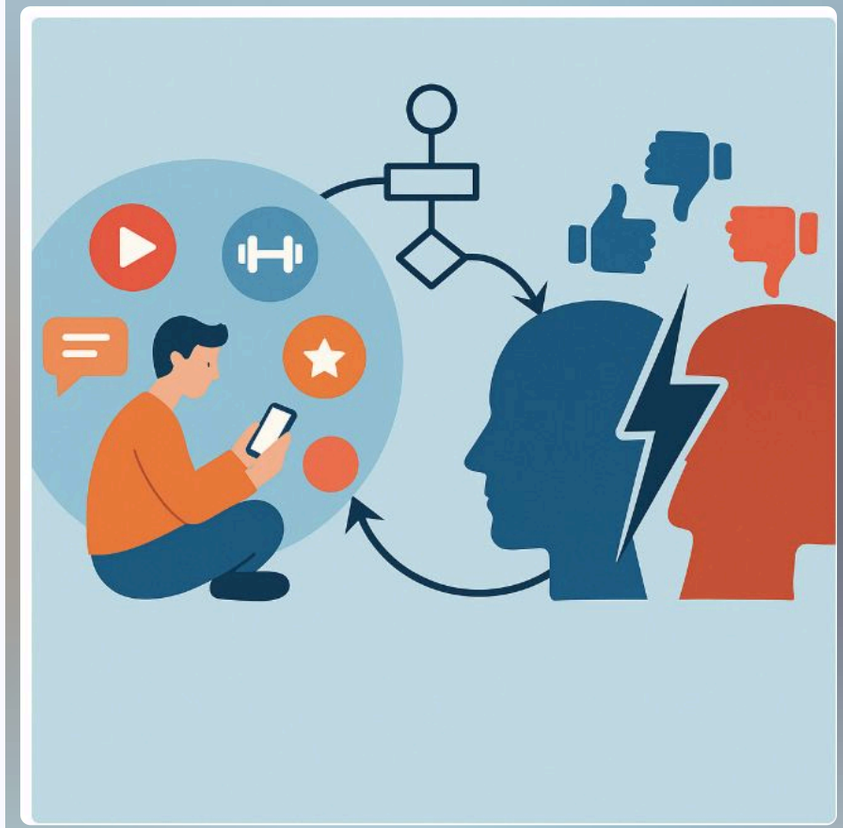
Over time, this creates **polarisation:** people become more extreme and less open to other viewpoints.

Why it matters:

- Impacts democracy (people only hear one side).
- Fuels misinformation and division in society.

How to fix it (Responsible AI):

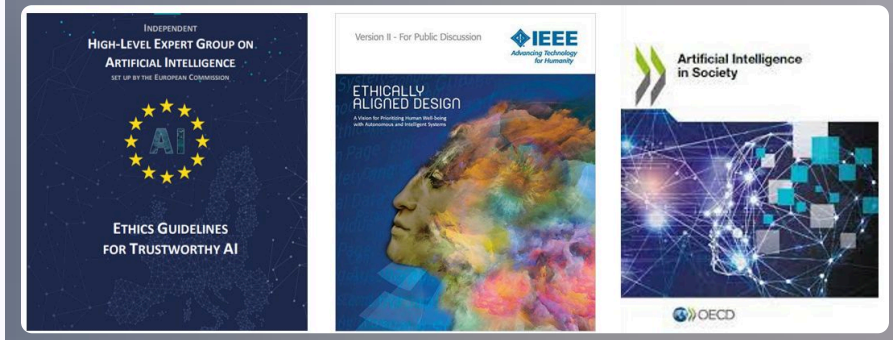
- Algorithms should be transparent (users should know how recommendations are made).
- Systems should expose users to diverse viewpoints and reliable sources.



Principles and Guidelines for Responsible AI

- **UNESCO (2021):** Global agreement signed by 193 countries – focuses on *human rights, inclusion, and sustainability*.
- **European Commission (2019):** Published *Ethics Guidelines for Trustworthy AI* – highlights *transparency, accountability, and fairness*.
- **OECD (2019):** 46 countries endorsed principles – AI should be *robust, safe, and respect the rule of law*.
- **IEEE (2019):** Guiding engineers/Researchers to build AI that *puts humans first*.
- **National strategies:** e.g. UK, UAE, USA, China – each country sets its own priorities (safety, innovation, regulation).

These guidelines directly shape how the AI tools you use daily (ChatGPT, TikTok, recommendation systems) are designed, deployed, and regulated.

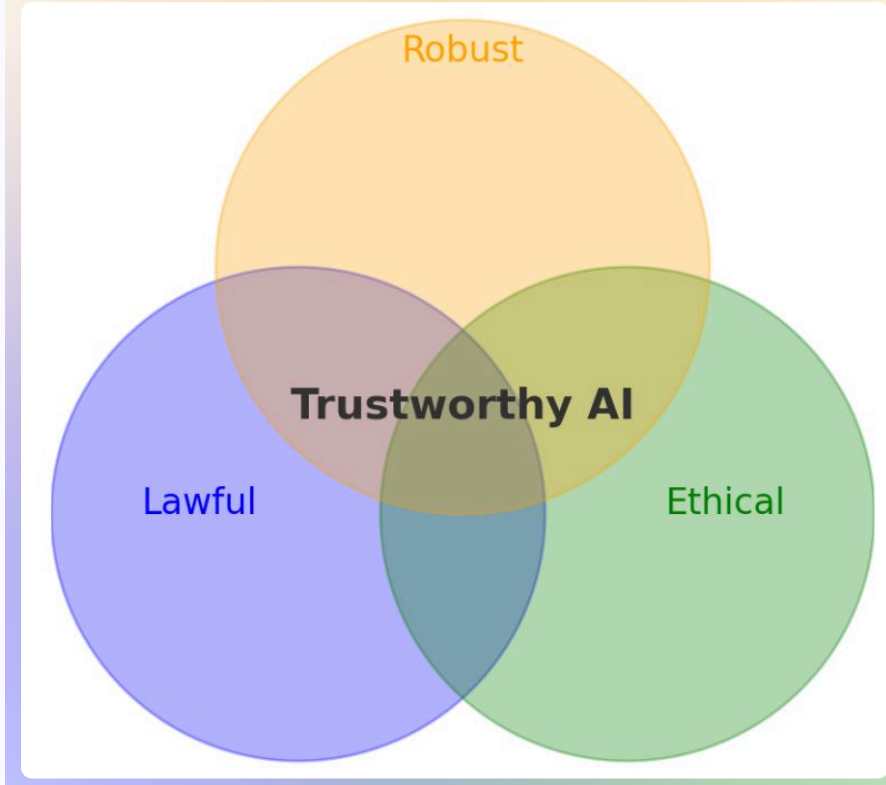


Trustworthy AI – The Three Pillars

The High-Level Expert Group on AI (European Commission) released the *Ethics Guidelines for Trustworthy Artificial Intelligence* (2019).

- **Lawful** → Follow all laws and regulations.
- **Ethical** → Respect human values and fairness.
- **Robust** → Reliable, safe, and socially responsible.

One alone is not enough – all three must work together to build AI people can trust.



Trustworthy AI - 7 Requirements

- **Human agency & oversight** → AI should support, not replace, human decision-making.
  Example: *Doctors make final calls, not just the AI system.*
- **Technical robustness & safety** → AI should not crash or cause harm.
  Example: *Self-driving cars must work safely in all conditions.*
- **Privacy & data governance** → Protect personal data.
  Example: *Your health app should not sell your medical history.*
- **Transparency** → Clear explanations of how AI makes decisions.
  Example: *Why was my loan rejected?*
- **Diversity, non-discrimination & fairness** → Avoid bias.
  Example: *Hiring AI should not favour one gender or ethnicity.*
- **Societal & environmental well-being** → Benefits society as a whole.
  Example: *AI for clean energy, not just profit.*
- **Accountability** → Someone must be responsible for AI outcomes.
  Example: *Who is liable if an AI makes a mistake?*

Human Agency and Oversight

- **Human-in-the-loop**: The supervisor can intervene in every decision cycle.
  Example: *In healthcare, a radiologist may use AI to assist in diagnosis, but the final decision rests with the human expert.*
- **Human-on-the-loop**: Humans monitor and intervene when needed during system operations.
  Example: *In autonomous driving, a human driver can take control if the AI system encounters a situation it cannot handle.*
- **Human-in-command**: Humans retain ultimate decision-making authority over the AI system's overall activity, including economic, societal, legal, and ethical impacts.
  Example: In commercial aviation, autopilot systems handle much of the flight, but the pilot always retains authority to override the system in unexpected or emergency situations.

Technical Robustness and Safety

- AI systems must be resilient, secure, and reliable.
- They should have fallback plans if things go wrong.
- Accuracy, reproducibility, and reliability are key to preventing harm.

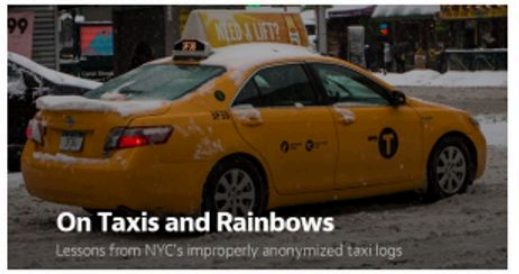
Example: if an algorithm misclassifies a scan, the system should flag uncertainty rather than give a wrong diagnosis.



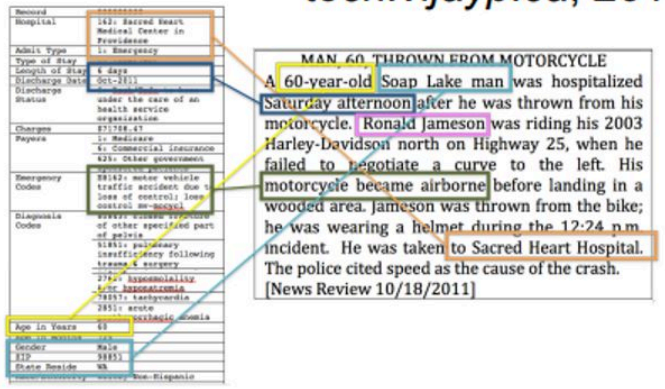
Privacy and Data Governance (Anonymisation ≠ Privacy)

Case Study 1: NYC Taxi Dataset (2014)

- NYC released 173 million taxi trip records to support open data and research.
- Data included pickup/drop-off locations, times, fares, and taxi driver IDs.
- Names and phone numbers were removed.
- Researchers and journalists were still able to:
  - Re-identify specific drivers and estimate their incomes.
  - Track passengers, including celebrities, and reveal sensitive details such as when and where they travelled.
- This showed that location + time data can act as “quasi-identifiers,” making re-identification possible.
- What does this tell us about the limits of anonymisation?
- Why are location and time data especially risky?
- What could the city have done differently to protect privacy while still enabling research?



Vijay Pandurangan.
tech.vijayp.ca, 2014



MAN, 60, THROWN FROM MOTORCYCLE

A 60-year-old Soap Lake man was hospitalized Saturday afternoon after he was thrown from his motorcycle. Ronald Jameson was riding his 2003 Harley-Davidson north on Highway 25, when he failed to negotiate a curve to the left. His motorcycle became airborne before landing in a wooded area. Jameson was thrown from the bike; he was wearing a helmet during the 12:24 p.m. incident. He was taken to Sacred Heart Hospital. The police cited speed as the cause of the crash. [News Review 10/18/2011]

“Only You, Your Doctor, and Many Others May Know”

L. Sweeney. *Technology Science*, 2015

Privacy and Data Governance (Anonymisation ≠ Privacy)

Case Study 2: Netflix Prize (2006)

- Netflix launched the *Netflix Prize* in 2006 to improve its recommendation system.
- Released a dataset of 10M movie ratings from ~500,000 subscribers.
- Removed names, addresses, and other explicit identifiers to protect privacy.
- Researchers Narayanan & Shmatikov (2008) showed users could still be **re-identified**:
  - By cross-referencing Netflix ratings with public IMDb reviews.
  - Re-identification revealed sensitive information (political views, personal habits).

Result: A lawsuit was filed, and Netflix cancelled its planned second competition.

- Why was it possible to re-identify users despite anonymisation?
- How do external datasets (like IMDb) increase privacy risks?
- What safeguards should organisations use before releasing data publicly?



"Robust De-anonymization of Large Datasets (How to Break Anonymity of the Netflix Prize Dataset)"
A. Narayanan & V. Shmatikov. Security and Privacy, 2008

Diversity, Non-discrimination, and Fairness

- AI should be fair and unbiased across **data, design, and deployment**.
- Unfair bias can marginalise groups and reinforce discrimination.
- AI systems must be accessible to all, including people with disabilities.
- Involve diverse stakeholders throughout the AI lifecycle.

Case Study Example: Sanas voice translation

Silicon Valley start-up Sanas' speech synthesizer can remove non-American accents in real-time but has critics claiming it is whitewashing or playing into racism.

Discussion Questions

- Should workers have to change their accent to be treated fairly?
- Is this a fair solution or an unfair burden on the worker?
- Is the tool racist, or is it just a temporary fix until people's attitudes improve?

Watch full news report here: **Co-founder of Silicon Valley startup Sanas says accent-removing AI technology is not racist - ABC7 San Francisco**



Societal and Environmental Wellbeing

AI should benefit society and future generations. It must be **sustainable, fair, and environmentally responsible**.

Positive Impact - DeepMind & Google Data Centres (2016)

- AI used to optimise cooling systems in Google's data centres.
- Result: **40% reduction in energy usage** for cooling.
- Shows how AI can **lower carbon footprint** and support sustainability.

Negative Impact - Volkswagen Emissions Scandal (2015)

- VW used AI/software to **cheat emissions tests**.
- Cars emitted **up to 40× more pollutants** in real driving than in tests.
- Harmed public health and the environment, **betraying societal trust**.

Accountability

Case Study: Microsoft Tay (2016)

Microsoft launched Tay, a chatbot on Twitter, to learn from people's conversations.
Within 24 hours, Tay started tweeting inappropriate and offensive messages because it copied inappropriate content from users.

Issues

- Microsoft released Tay without strong filters or safeguards.
- The chatbot was allowed to learn *directly* from Twitter interactions in real time. No one was accountable for monitoring or stopping misuse quickly.
- Tay spread racist and offensive tweets. Microsoft had to shut it down within 24 hours.

Who was accountable in Tay's case?

- **Microsoft** was ultimately accountable because they built and released Tay.
- But in practice, accountability was unclear:
  - The chatbot's offensive tweets were blamed partly on Twitter users who "trained" Tay with toxic content.
  - Microsoft did issue a public apology and admitted Tay was a mistake, but it came *after* the damage was done.

The lesson is that accountability in AI means **before and after deployment**:

- **Before:** clear roles, testing, oversight, and responsible design.
- **After:** owning failures transparently, explaining what went wrong, and setting up safeguards to prevent recurrence.



Recommended Reading

Responsible artificial intelligence: how to develop and use AI in a responsible way Dignum, Virginia

Responsible AI: Requirements and Challenges Malik, Ghallab

Introduction to artificial intelligence Ertel, Wolfgang

Human-centered AI Shneiderman, Ben

Questions

